

Region-wise BrainAGE Prediction

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I. INTRODUCTION

Neurodegenerative diseases, such as Alzheimer’s disease, cause the brain to change in distinct ways compared to healthy individuals. One of the keys to identify such behavior is to analyse these changes to improve the early detection and identification of neuro pathologies. In this context, Brain Age (BA) is a major biomarker for neurological research. Its estimation enables a comparison between the apparent age of the brain, and its real age, a large gap suggesting neurological deviation. For this analysis to be accurate and informative, the prediction method must be precise and interpretable.

Most of previous works estimated a single value from the whole brain volume. Nguyen et al. [8], among other works, split the MRI into 125 overlapping sub-volumes, allowing the model to build its representation and features from different zones of the brain that may follow different aging patterns. However, these sub-volumes are not anatomically grounded, limiting the model’s ability to associate predictions with specific brain structures.

In this work, we provide a more precise atlas-based parcellation for the regions of interest (ROI) the model considers. Neurodegenerative diseases tend to show different effects based on the ROI considered — for instance, the hippocampus undergoes substantial atrophy in the progression of Alzheimer’s disease [1]. We therefore propose a framework that extracts a fixed-size feature representation for each of 18 literature-grounded ROIs via ROI pooling on hierarchical CNN features, and encodes them through a Transformer to produce both per-ROI and subject-level age predictions. Our hypothesis is that this neuroanatomically grounded approach improves both prediction accuracy and interpretability, enabling a structured comparison of regional brain aging patterns across subjects.

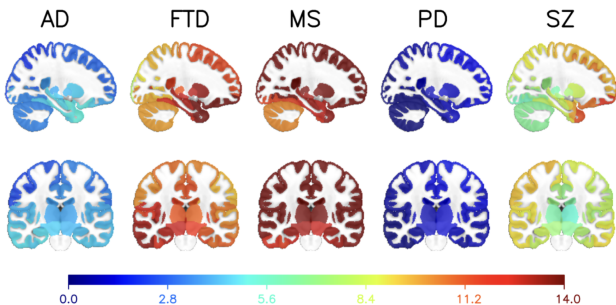


Fig. 1. BSAGE of different populations - Nguyen et. al. [8]

II. RELATED WORK

A. Global Brain Age Prediction

Neuroimaging biometric brain age prediction enables assessment of brain structure differences between individuals and standard aging patterns. Franke et al. demonstrated that healthy adult chronological age estimation through T1-weighted MRI required kernel-based machine learning models which they tested during their initial research work [2]. The brain age gap (BAG) which measures the difference between predicted age and actual chronological age has become an established brain health marker that shows increased BAG values link to neurodegenerative diseases and cognitive impairment and higher death rates [3]. CNNs lead deep learning methods to decrease prediction errors until they reach minimum average errors of 2–4 years because they use less processed data for prediction analysis [4].

Current brain age models are typically based on a single scalar value, which limits their ability to capture region-specific aging patterns. [5]. Major neurological disorders which require specific brain circuits for their development cannot be tracked through a global index since each disorder displays unique spatial patterns of brain progression [5], [6]. Whitmore and Beck argued that globally averaged estimates obscure region-specific maturational delays especially in developmental populations [7].

B. Spatially Resolved Brain Age Mapping

Recent studies have proposed spatial brain age mapping approaches. [8] created a voxel-based age mapping system that uses multiple deep learning models to produce *Brain Structure Ages* which provide better performance than standard scalar BAG methods when diagnosing Alzheimer’s disease, Parkinson’s disease, FrontoTemporal Dementia and schizophrenia.

Gianchandani et al. [5] developed a multitask U-Net system which creates per-voxel PAD maps while conducting tissue segmentation and determining global age. The system shows developed patterns of health deterioration through spatially spread patterns of health deterioration in the ventricles and caudate and temporal lobe which the system measured through standardized methods that allowed objective assessment among different groups of people. Riccardi et al. [6] discovered six different brain aging patterns occurring throughout the brain of healthy adults, showing that specific brain regions age differently to account for differences in mental performance and behavioral results. Sanford et al. [9] demonstrated that BrainAGE measurement shows different patterns based on sex, as women show increased brain aging

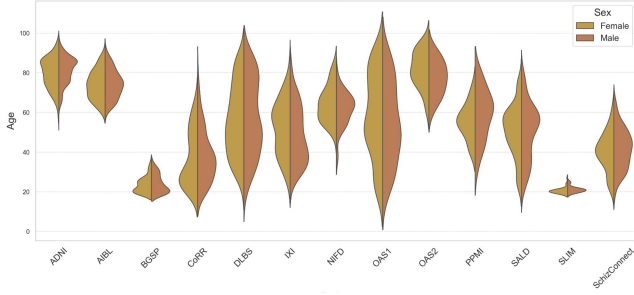


Fig. 2. Violin plot of ages in dataset sub-parts

in their cerebellum and brainstem areas. This research shows the importance of controlling for demographic factors when studying brain function at specific regional areas.

III. MATERIAL AND METHODS

A. Dataset and preprocessing

We use a multi-cohort dataset of 2,851 T1-weighted MRI scans from public datasets (ADNI, AIBL, OASIS, IXI, SALD, DLBS, CoRR, SchizConnect, NFD, PPMI, BGSP), covering a wide age range and multiple clinical populations. The data is split at the subject level into train/validation/test sets with balanced age and cohort distributions to avoid leakage and site bias. All images are processed through a standardized pipeline (skull stripping, registration to a common template, intensity normalization). We account for multi-site variability and will evaluate the impact of scanner effects. The main task is age regression, evaluated both globally and at the regional level.

MRI derived modalities include FAST GM/WM/CSF probability maps, deformation maps by ANTs, warps / inverse-warp, Jacobian determinant and Freesurfer Segmentation map (all in template spaces).

B. Problem Formulation

Let $\mathbf{V} \in \mathbb{R}^{1 \times D \times H \times W}$ denote a skull-stripped, affinely registered T1-weighted MRI volume and $y \in \mathbb{R}^+$ the corresponding chronological age. A paired anatomical label map $\mathbf{L} \in \{0, \dots, K\}^{D \times H \times W}$, derived from FreeSurfer `aparc+aseg+hsf` segmentation [16], assigns each voxel to one of $K = 18$ ROIs selected on the basis of established aging literature (Table I).

The model is trained to produce a subject-level prediction $\hat{y}_{\text{subj}} \in \mathbb{R}$ and a set of ROI-level predictions $\{\hat{y}_k\}_{k=1}^K$, where $\hat{y}_k$ reflects the age estimated from the structural features of ROI k alone. The regional brain age gap is then defined as

$$\Delta_k = \hat{y}_k - y. \quad (1)$$

C. ROI Selection

We select 18 bilateral ROIs (Table I) whose age-related structural changes are well-established in the literature. Subcortical structures (hippocampus, amygdala, lateral ventricle, thalamus, caudate, putamen) are chosen on the basis of documented volumetric atrophy or expansion with aging [10]–[12], [14]. Cortical regions (superior frontal, middle temporal,

insula) are selected following evidence of differential cortical thinning [13], [15]. An ROI is included only if its resampled voxel count exceeds 64, ensuring that even compact structures such as the amygdala yield a meaningful pooled representation.

TABLE I
ANATOMICAL ROIS AND SUPPORTING REFERENCES

Region	FreeSurfer ID (L/R)	Reference
Hippocampus	17/53	[10], [11]
Amygdala	18/54	[15]
Lateral Ventricle	4/43	[12]
Thalamus	10/49	[14]
Caudate	11/50	[14]
Putamen	12/51	[14]
Superior Frontal	1028/2028	[13]
Middle Temporal	1015/2015	[13], [15]
Insula	1035/2035	[15]

D. Model Architecture

The proposed model comprises three sequential modules: a hierarchical 3D CNN encoder, an ROI feature pooling module, and a Transformer-based aggregation network.

1) *Hierarchical 3D CNN Encoder*: We adopt the encoding pathway of a U-Net architecture [17] as a multi-scale feature extractor, retaining only the contracting path and discarding the decoder and skip-connection fusion. This design choice is deliberate: decoding to full resolution is unnecessary when the downstream objective is feature extraction at the ROI level. By omitting the decoder, the encoder operates as a hierarchical feature pyramid that preserves spatial coordinates at each downsampling level—a prerequisite for spatially grounded ROI pooling.

The encoder consists of five convolutional stages with progressively increasing channel dimensions. Each stage applies two successive $3 \times 3 \times 3$ convolutions with batch normalization and ReLU activation (DoubleConv), followed by $2 \times 2 \times 2$ max-pooling, yielding a four-level feature hierarchy in which each level operates at half the spatial resolution of the preceding one.

The level-3 feature map ($8 \times$ downsampled) is selected as the pooling target. This resolution represents a principled balance between spatial specificity—sufficient to resolve structural differences within compact nuclei such as the hippocampus—and semantic richness accumulated through four stages of nonlinear feature extraction. Dropout ($p = 0.1$) is applied after each DoubleConv block.

2) *ROI Feature Pooling*: For each ROI k , a binary mask $\mathbf{m}_k \in \{0, 1\}^{20 \times 24 \times 20}$ is obtained by nearest-neighbor downsampling of the label map $\mathbf{L}$ to the level-3 feature map resolution. The non-zero voxel coordinates of $\mathbf{m}_k$ define a three-dimensional bounding box, within which the feature map is cropped:

$$\mathbf{c}_k = \mathbf{F}_{\ell=3} [d_{\min}^k : d_{\max}^k, h_{\min}^k : h_{\max}^k, w_{\min}^k : w_{\max}^k], \quad (2)$$

where $\mathbf{F}_{\ell=3} \in \mathbb{R}^{256 \times 20 \times 24 \times 20}$. The crop $\mathbf{c}_k$ is resampled to a fixed resolution of $4 \times 4 \times 4$ via trilinear interpolation, yielding

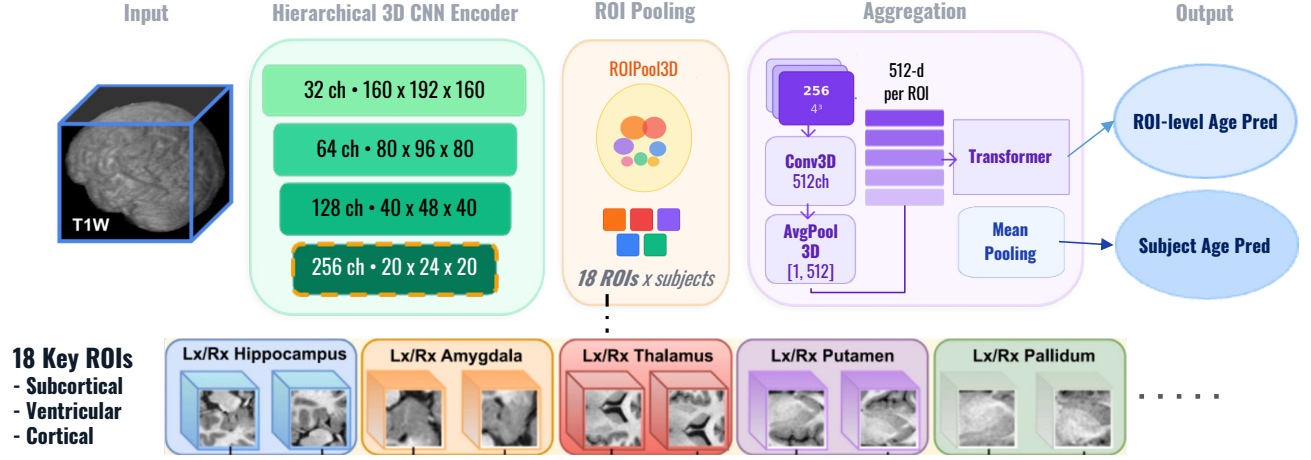


Fig. 3. Proposed pipeline

$\mathbf{r}_k \in \mathbb{R}^{256 \times 4 \times 4 \times 4}$ irrespective of the native spatial extent of ROI k . This procedure is analogous to RoIAlign [18] in the volumetric setting, and ensures that both large cortical regions (e.g., superior frontal, $\approx 35,000$ voxels) and compact subcortical nuclei (e.g., amygdala, $\approx 2,000$ voxels) contribute at uniform dimensionality.

Each pooled representation is encoded by a *shared* ROI feature extractor:

$$\mathbf{f}_k = \text{FC}(\text{AvgPool3D}(\text{Conv3D}^{\times 2}(\mathbf{r}_k))) \in \mathbb{R}^{512}, \quad (3)$$

where the two $3 \times 3 \times 3$ convolutional layers double the channel dimension to 512, global average pooling removes spatial axes, and the fully connected layer projects to 512 dimensions. Dropout ($p = 0.1$) follows the projection. The shared encoder weights are applied identically to all 18 ROIs; region-specific differences are consequently learned by the subsequent Transformer and prediction heads.

3) *Transformer Aggregation*: The sequence of $K = 18$ ROI embeddings $\mathbf{F}^{\text{roi}} = [\mathbf{f}_1, \dots, \mathbf{f}_K] \in \mathbb{R}^{K \times 512}$ is processed by a Transformer encoder [19] with $L = 2$ layers, 4 attention heads, feed-forward expansion ratio 4.0, and dropout $p = 0.1$. Self-attention across the 18 ROI tokens enables the model to capture inter-regional co-aging dependencies—for example, the coordinated atrophy of the hippocampus and expansion of the lateral ventricles that accompanies normal aging and is accentuated in Alzheimer’s disease [10], [14].

Following the Transformer, mean pooling collapses the token sequence to a single subject-level representation:

$$\mathbf{g} = \frac{1}{K} \sum_{k=1}^K \tilde{\mathbf{f}}_k \in \mathbb{R}^{512}, \quad (4)$$

where $\tilde{\mathbf{f}}_k$ is the contextualized embedding of ROI k from the final Transformer layer.

4) *Prediction Heads*: **ROI-level head**. A shared linear layer with dropout ($p = 0.1$) is applied independently to each contextualized ROI embedding:

$$\hat{y}_k = \mathbf{w}_{\text{roi}}^T \text{Dropout}(\tilde{\mathbf{f}}_k) + b_{\text{roi}}, \quad k = 1, \dots, K. \quad (5)$$

Because each ROI prediction receives an independent gradient during backpropagation, the model is incentivized to develop region-discriminative features. This end-to-end ROI supervision is the principal distinction from post-hoc aggregation approaches [8], where structure-level ages are derived by averaging voxel-wise predictions over a separately trained segmentation model.

Subject-level head. A separate linear layer with dropout ($p = 0.2$) maps the global representation to a scalar age:

$$\hat{y}_{\text{subj}} = \mathbf{w}_{\text{subj}}^T \text{Dropout}(\mathbf{g}) + b_{\text{subj}}. \quad (6)$$

This head integrates information from all 18 ROIs through the Transformer’s contextualized mean pooling, and serves as the primary performance metric.

E. Training Objective

The model is trained with a composite multi-task loss:

$$\mathcal{L} = \lambda_{\text{subj}} \mathcal{L}_{\text{subj}} + \lambda_{\text{roi}} \mathcal{L}_{\text{roi}}, \quad (7)$$

with $\lambda_{\text{subj}} = \lambda_{\text{roi}} = 1.0$.

Subject-level MSE. The subject head is supervised directly against chronological age:

$$\mathcal{L}_{\text{subj}} = \frac{1}{B} \sum_{i=1}^B (\hat{y}_{\text{subj}}^{(i)} - y^{(i)})^2. \quad (8)$$

ROI-level MSE. Each ROI prediction is independently supervised against the same chronological age label, providing direct gradient feedback to every region:

$$\mathcal{L}_{\text{roi}} = \frac{1}{B} \sum_{i=1}^B \frac{1}{K_i} \sum_{k=1}^{K_i} (\hat{y}_k^{(i)} - y^{(i)})^2, \quad (9)$$

where K_i is the number of valid ROIs for subject i . This per-ROI supervision is what drives region-specific interpretation: by receiving independent gradient signals, each region learns to extract age-discriminative features from its own structural representation, rather than relying on a globally averaged signal.

F. Experimental Setups

The model was trained for 30 epochs using the AdamW optimizer [20] with a learning rate of 2×10^{-4} and weight decay of 10^{-4} . The batch size was set to 4, with gradient accumulation over 4 steps yielding an effective batch size of 16.

IV. RESULTS

Table II reports brain age estimation performance on the held-out test set of 424 subjects, alongside the baseline method of Nguyen et al. [8] for reference.

TABLE II
TEST SET PERFORMANCE

Method	Level	MAE (yr)	R^2
Nguyen et al. [8]	Subject (young / old)	1.88 / 3.83	0.91 / 0.61
	Regional	—	—
Proposed	Subject	8.24	0.89
	ROI (avg.)	9.37	0.86

The proposed model achieves a global MAE of **8.24 years** and $R^2 = 0.89$ at the subject level, indicating strong agreement between predicted and chronological age across the full lifespan range of 18–91 years. At the ROI level, the model yields an MAE of **9.37 years** and $R^2 = 0.86$, demonstrating that region-specific predictions retain predictive validity comparable to the global estimate. The 1.13-year gap between the two levels is expected: the subject head integrates contextualized features from all 18 ROIs through Transformer-mediated mean pooling, whereas each ROI head operates on a single region’s representation and is subject to the informational limits of that region’s age-discriminability.

The primary distinction from Nguyen et al. [8] lies in how regional predictions are derived. Their approach obtains structure-level ages as a post-hoc average of voxel-wise predictions from an ensemble of 125 U-Nets, where the decoder is trained solely on global reconstruction objectives. In contrast, the proposed model learns ROI-discriminative representations end-to-end: each of the 18 regions receives independent gradient supervision through $\mathcal{L}_{\text{roi}}$, directly incentivizing the encoder to produce features that are informative at the regional level. This architectural choice, in exchange for a more complex optimization problem, enables interpretable per-ROI brain age gap profiles without any post-hoc processing.

V. CONCLUSION

We experimented a region-wise Brain Age framework via ROI pooling on hierarchical CNN features. This framework enables per-ROI age deviation alongside subject-level prediction. The global MAE of 8.24 yrs ($R^2=0.89$) shows consistent ROI-level predictions across 18 anatomical regions. Despite not reaching SOTA prediction accuracies, our model show that a network designed from anatomical reasoning to prioritize interpretability and can still reach decent accuracy.

Future Directions:

One current limitation is that our model is a single-network, and could benefit from complexity optimization and modified scale. A systematic ablation study should also be performed to assess the influence of different ROIs, as it could improve the model performance, complexity and interpretability. A promising future direction is also the addition of multimodal inputs. The preprocessing of the dataset provides deformation fields to assess displacement contribution, which can be informative on atypical neurological behavior. Finally, external validation should be performed on independent cohorts such as UK Biobank’s dataset.

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